

MODULE 26 · PART 8 - PRACTICE

What Holds

Twelve hours, twenty-five modules, and about seven ideas.

PART 8 of 8 MODULE 26 of 26 RUNS about 15 min

You will forget most of this. **That is the normal outcome**, and it is not the failure it looks like.

What goes, and what stays

- ◆ **Syntax goes.** Frame keywords, function names, the dialect table from module fifteen. All of it, quickly.
- ◆ **Specific numbers go.** Nobody needs to remember that 31% of Kestrel's transactions tie within a second.
- ◆ **Habits stay,** because they are things you do rather than things you know.
- ◆ **And a changed default stays longest of all,** because it costs nothing to maintain.

THE TAKEAWAY

The measure of this course is not what you can recall. It is whether you now write a tiebreaker without deciding to, which is a different kind of change.

There were **about seven ideas** here.
Everything else was one of them, wearing a
different clause.

The seven

THE IDEA	WHERE IT KEPT APPEARING
NULL is three claims and one value	02, 06, 07, 09, 15, 17, 18, 19
State what one row means	01, 04, 10, 13, 15, 17, 20
An incomplete ordering is not an ordering	02, 04, 05, 13, 17
A denominator is a population	07, 08, 09, 14, 19
Absence is invisible	06, 12, 13, 15, 17
Compute the shape the questions want, once	10, 13, 14, 15, 22
Write the belief where it can be contradicted	20, and every module's last slide

THE TAKEAWAY

Seven rows, and the module numbers are the evidence. A thing appearing in five modules under five names was never about those five topics.

NULL is **three different claims** and SQL gives you one value for all of them.

How it kept showing up

- ◆ **Comparing versus collecting.** A join says two nulls cannot be shown equal; a GROUP BY and a set operation say they are the same.
- ◆ **Skipped versus counted.** AVG has a denominator you cannot see, and COUNT of a column is a different question from COUNT of a star.
- ◆ **Absent versus recorded-as-nothing.** An empty array and a null array both produce no rows, and mean different things.
- ◆ **And invented versus real.** A subtotal null and a data null share a column, and only GROUPING separates them.

THE TAKEAWAY

Four faces of one idea, spread across eight modules. Each one is a place where SQL had to pick a behaviour and picked a reasonable one that is not always what you meant.

State what one row means. Half of everything else follows from that sentence.

What the sentence gives you

- ◆ **The grain assertion writes itself.** Row count equals distinct count of the key you just named.
- ◆ **Fan-outs become visible,** because a join that changes the sentence has changed the table.
- ◆ **And a stacked query documents itself,** one grain per block, which is module ten's shape and module twenty-three's bisection.
- ◆ **Most tables cannot answer it,** which is itself the finding, and it takes ten seconds to discover.

THE TAKEAWAY

Ask a team what one row of their largest table means. The hesitation before the answer is proportional to the number of defects in everything built on it.

An ordering that does not **uniquely identify a row** is not an ordering. It is a set of permitted answers.

Where it matters, and where it does not

- ◆ **It matters wherever the ordering decides something.** A ROW_NUMBER, a LIMIT, a frame boundary, a first or last value.
- ◆ **It does not matter for presentation,** where an ORDER BY at the very end is arranging rows nobody will cut.
- ◆ **The test is whether a tie changes the answer,** rather than the order the answer arrives in.
- ◆ **And running it twice can fail you but cannot clear you,** which module twenty was careful about.

THE TAKEAWAY

The distinction is worth keeping: an ordering that only sorts output is free to be incomplete. An ordering that selects, ranks or bounds is not.

A denominator is **a population**. Almost nobody checks which one.

The check that actually works

- ◆ **Shares summing to one proves nothing**, because a denominator covering any population at all will satisfy it.
- ◆ **Two levels multiplying out does prove something**, because a filtered denominator makes the levels disagree.
- ◆ **And an index averaging to a hundred is a third check**, catching an average over the wrong partition.
- ◆ **One line each.** The weak one is the one everybody writes, and it is satisfied by the defect.

THE TAKEAWAY

This is the single most transferable assertion in the course. Any report with a share at two levels has the property, and checking it costs one row of output.

Absence is invisible. Nothing in a result tells you what is not in it.

Which is why you count

- ◆ **Row count in against row count out**, which is one comparison and catches both loss and fan-out at once.
- ◆ **Coverage as arithmetic rather than equality**, so the difference is explained rather than merely absent.
- ◆ **Because nothing else will tell you**. No error, no warning, and usually a total that looks about right.
- ◆ **And absence at the edges is doubly invisible**, which is why gaps are measured against a declared window rather than against the data.

THE TAKEAWAY

Every silent-loss defect in this course would have been caught by one comparison of two row counts. It is the cheapest assertion and the one most often absent.

Compute the shape the questions want,
once.

And the reason is not performance

- ◆ **Performance is the visible argument**, and it is usually the smaller one. Measured here, often a factor of five or seven.
- ◆ **Consistency is the real one**. Eleven definitions of a session become one, and stop drifting apart.
- ◆ **And the parameter becomes visible**. A threshold in one named place is a decision; in eleven places it is folklore.
- ◆ **When both arguments point the same way, take it**. That happens rarely and it happened in modules 13, 14 and 15.

THE TAKEAWAY

A materialised spell table is faster and it is not why you build it. You build it so that six reports cannot disagree about when a store was trading.

An assertion is a belief, written where it can be contradicted.

What that buys, in order of value

- ◆ **Least:** today's output is **probably fine**, which is what people think assertions are for.
- ◆ **More:** a reviewer learns **what the query is for**, in thirty seconds, without reading it.
- ◆ **More still:** a successor inherits the **beliefs**, not just the code, which is the only documentation that fails when it stops being true.
- ◆ **Most:** a change that **breaks an assumption fails loudly**, rather than being found by finance a quarter later.

THE TAKEAWAY

Three of the four are about other people and about the future. Almost none of the value of an assertion suite is about the run you are doing now.

This course covered **one layer** of a large subject, and left out a great deal on purpose.

What is not in here

- ◆ **Physical design.** Distribution keys, sort keys, clustering strategy, file layout. Touched in module twenty-one and not taught.
- ◆ **Streaming SQL,** where windows are over unbounded input and watermarks mean something different from module twenty-two's.
- ◆ **Geospatial, full-text, and vector search,** each of which is its own subject with its own operators.
- ◆ **And orchestration.** Module twenty-two discussed incrementality and said nothing about how a pipeline is scheduled or tested.

THE TAKEAWAY

Each of these is a course. The one thing here that transfers to all of them is the assertion discipline, because every one of them can be silently wrong too.

I recorded modules eleven and twelve with
a factual error in them, and found it by
building the data.

How I found it, and why it matters here

- ◆ **By building the data rather than describing it.** The cycle went into the capstone database, and the walk did not hang.
- ◆ **I would not have found it by reasoning,** because my reasoning was what produced the error in the first place.
- ◆ **Which is this whole course's argument, applied to itself.** A belief I held confidently, contradicted by running it.
- ◆ **So check me.** The database ships with the course precisely so that you can.

THE TAKEAWAY

The two modules were corrected before release. The reason for mentioning it is that it is the best example in the course of why you run the thing rather than reason about it.

Which is how to read any course like this

- ◆ **Confident delivery is not evidence.** I have been confident for twelve hours, and I was wrong about at least one thing.
- ◆ **The measured claims are the load-bearing ones**, and every number in this course was measured against the shipped database.
- ◆ **The unmeasured ones are opinions**, including most of the advice about what you should do, and they are worth less.
- ◆ **And your engine's documentation beats me**, always, on anything about your engine.

THE TAKEAWAY

The distinction worth keeping: which claims came with a number and a way to reproduce them, and which came with a confident tone. Only the first kind is evidence.

And what will have dated

- ◆ **Every engine support table in this course.** Redshift gained three clauses in 2023 alone, and Oracle gained four in 21c.
- ◆ **Postgres will get IGNORE NULLS**, and probably row pattern recognition, and both were under discussion while this was recorded.
- ◆ **So check before designing around a gap**, because a missing feature is the least durable fact in here.
- ◆ **The behaviours will not date.** NULL, ordering, frames and grain are in the standard and have been for decades.

THE TAKEAWAY

The dialect tables are the perishable part and the semantics are not. When something here disagrees with your engine's documentation, the documentation is right.

None of this is difficult. **The difficulty is remembering to ask**, when nothing is prompting you.

What to actually do next

- ◆ **Pick your largest table and state its grain in one sentence.** Then assert it. Fifteen minutes.
- ◆ **Find one report with a share column,** and check whether its two levels multiply out. Ten minutes.
- ◆ **Aggregate your query history by normalised text, sorted by total time,** and look at what is at the top. Five minutes.
- ◆ **And measure the arrival lag on your largest incremental table,** against the window it is configured with.

THE TAKEAWAY

Four things, under an hour, none of which needs permission. Between them they usually find something, and finding one thing is what makes the habit stick.

If you have not done the capstone

- ◆ **Then you have watched a course about doing something**, which is a different experience from having done it.
- ◆ **It is a weekend, and the harness marks you**, on properties rather than values, so it cannot be passed by copying.
- ◆ **And the database is the durable artefact here**, because every defect in twenty-five modules is actually in it.
- ◆ **Keep it after you finish.** It is a better fixture than anything you will build for yourself.

THE TAKEAWAY

The slides will fade. A 170 MB database containing fourteen real defects at their true proportions is a thing you can point a query at in two years' time.

The seven, one more time

SAY IT AS

- | | |
|---|---|
| 1 | Which of the three nothings is this? |
| 2 | What does one row mean? |
| 3 | Does this ordering identify a row? |
| 4 | A denominator over which population? |
| 5 | What is not in this result? |
| 6 | Am I computing this more often than it changes? |
| 7 | How would I know if this were wrong? |

THE TAKEAWAY

Seven questions. If you get into the habit of asking the seventh, the other six tend to follow, because you cannot answer it without them.

The eight parts, and what each was really about

PART	UNDERNEATH IT
1 Foundations	Clause order, and the three kinds of nothing.
2 Windows	A frame is a population, and an ordering is a promise.
3 CTEs and recursion	Where a query can be read, and where a walk can be stopped.
4 Sequencing	Runs, sessions and intervals: one technique, three costumes.
5 Reshaping	Runtime rows, parse-time columns, and the loss in between.
6 Sets and grouping	Comparing versus collecting, and the two kinds of null.
7 Proof and performance	Right, then fast, then affordable, in that order.
8 Practice	Whether any of it is yours.

THE TAKEAWAY

Eight parts, and the seven ideas run through all of them. If a part seemed to be about syntax, it was the one idea underneath that was doing the work.

Almost every defect in this course was found by **somebody who was not looking for it.**

Thank you for staying for all of it. **Now go
and break something on purpose.**